

4.3.16

$$\frac{CD}{DB} = \frac{[CAD]}{[BAD]} = \frac{CA \sin \frac{A}{2}}{BA \sin \frac{A}{2}} = \frac{AC}{BA} = \frac{\sin B}{\sin C}$$

$$\frac{BF}{FA} = \frac{\sin \angle FCB}{\sin \angle FCA} = \frac{\sin A}{\sin B}$$

$$\text{and by Ceva's Theorem,}$$

$$\frac{\sin \angle FCB}{\sin \angle FCA} \cdot \frac{\sin \angle PAC}{\sin \angle PAB} \cdot \frac{\sin \angle PBA}{\sin \angle PBC} = 1$$

$$\frac{\sin \angle FCB}{\sin \angle FCA} \cdot \frac{\sin \frac{A}{2}}{\sin \frac{A}{2}} \cdot \frac{\cos \frac{A}{2}}{\sin(\frac{A}{2} + B)} = 1$$

$$\therefore \frac{BF}{FA} = \frac{\sin(\frac{A}{2} + B) \sin A}{\cos \frac{A}{2} \sin B}$$

$$\text{By symmetry } \frac{AE}{EC} = \frac{\cos \frac{A}{2} \sin C}{\sin(\frac{A}{2} + C) \sin A}$$

$$\therefore \frac{CD}{DB} \cdot \frac{BF}{FA} \cdot \frac{AE}{EC} = \frac{\sin B}{\sin C} \cdot \frac{\sin(\frac{A}{2} + B) \sin A}{\cos \frac{A}{2} \sin B} \cdot \frac{\cos \frac{A}{2} \sin C}{\sin(\frac{A}{2} + C) \sin A} = 1$$

We're done by Menelaus.

4.3.16)

